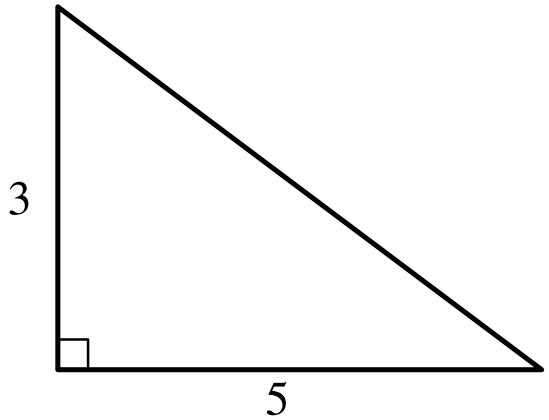
Area and volume — Part 2

Geometry and Trigonometry · Math · 55 questions

2.1 d341e9e5-5270-4fed-85ea-10d666d20e91 · Medium



Note: Figure not drawn to scale.

- One angle is a right angle.
- The lengths of the 2 sides adjacent to the right angle are as follows:
  - 3
  - 5
- A note indicates the figure is not drawn to scale.

The figure shows the lengths, in inches, of two sides of a right triangle. What is the area of the triangle, in square inches?

Student-produced response: _____

2.2 0baecf54-d44d-4084-ba8f-a8fb9f7596bf · Medium

A right circular cylinder has a height of 8 meters m and a base with a radius of 12 m. What is the volume, in m^3, of the cylinder?

A. 8π

B. 20π

C. 768π

D. $1{,}152\pi$

2.3 f60d8609-7495-41f1-a588-c1fdce04101d · Medium

A right circular cylinder has a base diameter of 22 centimeters and a height of 6 centimeters. What is the volume, in cubic centimeters, of the cylinder?

A. 132π

B. 264π

C. 726π

D. $2{,}904\pi$

2.4 d1c64540-1ef9-4449-9ccf-d158b23af654 · Medium

The length of each side of a square is 94 centimeters (cm). Which expression gives the area, in cm^2, of the square?

A. $2 \cdot 94$

B. $2 \cdot 94 \cdot 94$

C. $4 \cdot 94$

D. $94 \cdot 94$

2.5 0103767c-8ef7-4ec8-a9fb-3a3fa9a5469f · Medium

The length of one side of square M is 6 times the length of one side of square N. The area of square N is 49 square feet. What is the area, in square feet, of square M?

A. $1{,}764$

B. 504

C. 294

D. 252

2.6 eee80adc-71a4-4db9-8c03-1fddc462909d · Medium

A cylinder has a diameter of 8 inches and a height of 12 inches. What is the volume, in cubic inches, of the cylinder?

A. 16π

B. 96π

C. 192π

D. 768π

2.7 5b086c61-e308-4d87-b595-7b21a6d05163 · Medium

A hemisphere is half of a sphere. If a hemisphere has a radius of 27 inches, which of the following is closest to the volume, in cubic inches, of this hemisphere?

A. $1{,}500$

B. $6{,}100$

C. $30{,}900$

D. $41{,}200$

2.8 d0ab21dc-5c27-4da2-a45b-fa400982d6bc · Medium

A right circular cylinder has a volume of 432 cubic centimeters. The area of the base of the cylinder is 24 square centimeters. What is the height, in centimeters, of the cylinder?

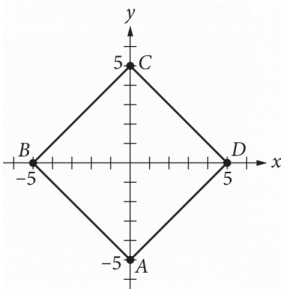
A. 18

B. 24

C. 216

D. 10,368

2.9 038021-DC · Medium



In the xy-plane shown, square ABCD has its diagonals on the x- and y-axes. What is the area, in square units, of the square?

A. 20

B. 25

C. 50

D. 100

2.10 b7b7672e-ce42-4e19-9094-c2ba6966c496 · Medium

A sphere has a radius of $\frac{17}{5}$ feet. What is the volume, in cubic feet, of the sphere?

A. $\frac{5\pi}{17}$

B. $\frac{68\pi}{15}$

C. $\frac{32\pi}{5}$

D. $\frac{19{,}652\pi}{375}$

2.11 9d90d9a2-0084-4cae-be11-73b040f11c9e · Medium

The length of the edge of the base of a right square prism is 6 units. The volume of the prism is $2{,}880$ cubic units. What is the height, in units, of the prism?

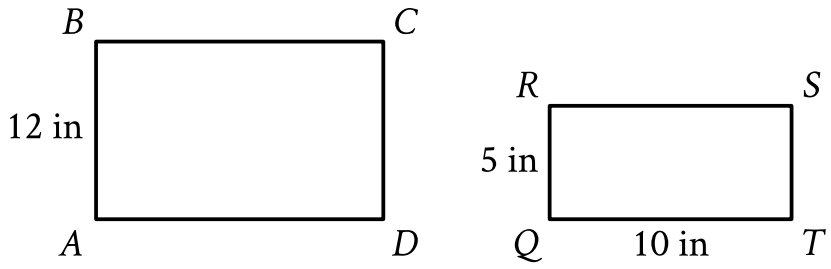
A. $4\sqrt{30}$

B. 36

C. $24\sqrt{5}$

D. 80

2.12 cc551413-74ed-4f97-8552-a6ef542a8584 · Medium



Note: Figure not drawn to scale.

- Rectangle upper A upper B upper C upper D is labeled as follows:
  - The length of side upper A upper B is 12 inches.
- Rectangle upper Q upper R upper S upper T is labeled as follows:
  - The length of side upper Q upper R is 5 inches.
  - The length of side upper Q upper T is 10 inches.
- A note indicates the figure is not drawn to scale.

Rectangles *ABCD* and *QRST* shown are similar, where *A*, *B*, *C*, and *D* correspond to *Q*, *R*, *S*, and *T*, respectively. What is the length, in inches in, of $\overline{AD}$?

A. 60

B. 24

C. 17

D. 10

2.13 fc7a142b-274d-4f6f-ab55-5f7708568898 · Medium

A circle has a radius of 43 meters. What is the area, in square meters, of the circle?

A. $\frac{43\pi}{2}$

B. 43π

C. 86π

D. $1{,}849\pi$

2.14 b609067c-22bb-404f-a813-9e197c6b487c · Medium

A cube has an edge length of 41 inches. What is the volume, in cubic inches, of the cube?

A. 164

B. 1,681

C. 10,086

D. 68,921

2.15 63912a12-3cfc-4144-bd81-5557667cbb81 · Medium

The table gives the perimeters of similar triangles TUV and XYZ, where TU corresponds to XY. The length of TU is 18.

	Perimeter
Triangle TUV	37
Triangle XYZ	333

What is the length of XY?

A. 2

B. 18

C. 55

D. 162

2.16 df0a6944-eb80-45dd-890a-e82089329b8e · Medium

A right circular cylinder has a height of 16 inches and a volume of $3{,}136\pi$ cubic inches. What is the radius, in inches, of this cylinder?

Student-produced response: _____

2.17 020533-DC · Medium

What is the volume, in cubic centimeters, of a right rectangular prism that has a length of 4 centimeters, a width of 9 centimeters, and a height of 10 centimeters?

Student-produced response: _____

2.18 027063-DC · Medium

A right circular cylinder has a volume of 45π. If the height of the cylinder is 5, what is the radius of the cylinder?

A. 3

B. 4.5

C. 9

D. 40

2.19 b6420961-9ecc-40b1-9203-ca168e783426 · Medium

The table gives the perimeters of similar triangles *TUV* and *XYZ*, where *TU* corresponds to *XY*. The length of *TU* is 6.

	Perimeter
Triangle *TUV*	50
Triangle *XYZ*	150

What is the length of *XY*?

A. 2

B. 6

C. 18

D. 56

2.20 aa19a0c6-76f9-4d66-b105-5e444ac3faec · Hard

A right circular cone has a volume of $71{,}148\pi$ cubic centimeters and the area of its base is $5{,}929\pi$ square centimeters. What is the slant height, in centimeters, of this cone?

A. 12

B. 36

C. 77

D. 85

2.21 205ab66a-8e60-4a19-9cbb-9c2754ce529f · Hard

The table gives the areas and perimeters of two similar rectangles, where n is a constant.

	Area (square inches)	Perimeter (inches)
Rectangle A	630	210
Rectangle B	2,520	n

What is the value of n?

A. 2,100

B. 1,680

C. 840

D. 420

2.22 c19a1158-3fde-4af0-be5d-79e3722ddf7e · Hard

A rectangular poster has an area of 360 square inches. A copy of the poster is made in which the length and width of the original poster are each increased by 20%. What is the area of the copy, in square inches?

Student-produced response: _____

2.23 25288de3-8156-4952-8e45-10506cfad636 · Hard

Rectangles $ABCD$ and $EFGH$ are similar. The length of each side of $EFGH$ is 6 times the length of the corresponding side of $ABCD$. The area of $ABCD$ is 54 square units. What is the area, in square units, of $EFGH$?

A. 9

B. 36

C. 324

D. $1{,}944$

2.24 395e8a55-7af9-456e-a433-3d4a89ade971 · Hard

A right rectangular prism has a length of 28 centimeters cm, a width of 15 cm, and a height of 16 cm. What is the surface area, in cm^2, of the right rectangular prism?

Student-produced response: _____

2.25 d6c651bd-349f-4a16-89ba-e8bb62e72076 · Hard

A right square pyramid has a total surface area of $70{,}560$ square inches, and the combined surface area of the four lateral faces of this pyramid is $38{,}160$ square inches. What is the height, in inches, of this pyramid?

A. 56

B. 90

C. 106

D. 180

2.26 131e158b-3eee-4f01-b713-1985fda38651 · Hard

The floor of a ballroom has an area of 600 square meters. An architect creates a scale model of the floor of the ballroom, where the length of each side of the model is $\frac{1}{10}$ times the length of the corresponding side of the actual floor of the ballroom. What is the area, in square meters, of the scale model?

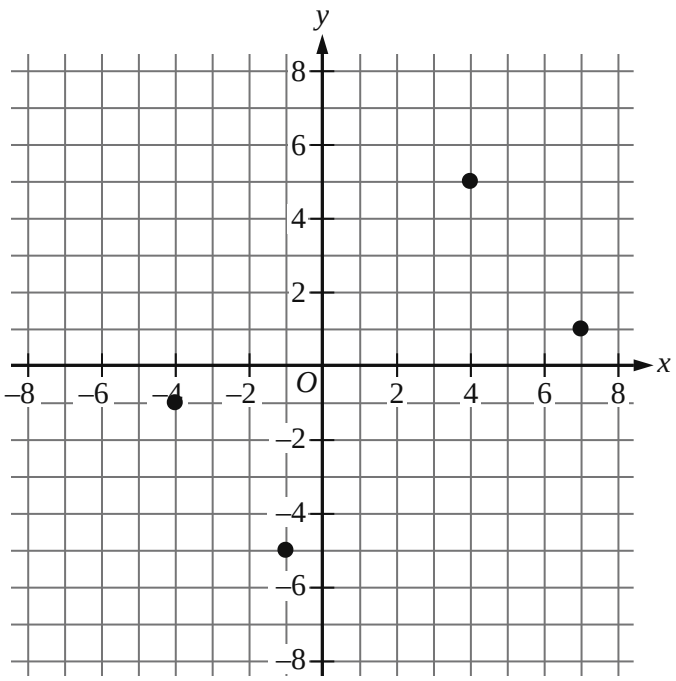
A. 6

B. 10

C. 60

D. 150

2.27 3dc2565f-6841-467d-aeb0-c0ac79807ad9 · Hard



- From left to right, the 4 points have the following coordinates:
  - (negative 4 comma negative 1)
  - (negative 1 comma negative 5)
  - (4 comma 5)
  - (7 comma 1)

A rectangle is formed by the four points shown. What is the area, in square units, of the rectangle?

Student-produced response: _____

2.28 70b95bcc-29ac-4b72-909d-c64fad3c9c1d · Hard

A right rectangular prism has a base area of $24t$ square centimeters cm^2. The length of the base of the rectangular prism is $\frac{8}{3}$ cm, and the height of the rectangular prism is 15 cm. Which expression represents the surface area, in cm^2, of the right rectangular prism?

A. $48t + 160$

B. $318t + 80$

C. $1{,}968t + 80$

D. $360t$

2.29 9a0d902b-5ffd-4b9f-99b5-e223fb690c35 · Hard

Right rectangular prism P is similar to right rectangular prism Q. The volume of prism P is 972 cubic centimeters cm^3, and the volume of prism Q is 36 cm^3. One edge of prism P has a length of $12\,\text{cm}$. What is the length, in cm, of the corresponding edge of prism Q?

Student-produced response: _____

2.30 05721-DC · Hard

The volume of right circular cylinder A is 22 cubic centimeters. What is the volume, in cubic centimeters, of a right circular cylinder with twice the radius and half the height of cylinder A?

A. 11

B. 22

C. 44

D. 66

2.31 ca49552d-a377-4804-b20d-7b730e15bb99 · Hard

Right rectangular prism X is similar to right rectangular prism Y. The surface area of right rectangular prism X is 58 square centimeters cm^2, and the surface area of right rectangular prism Y is $1{,}450$ cm^2. The volume of right rectangular prism Y is $1{,}250$ cubic centimeters cm^3. What is the sum of the volumes, in cm^3, of right rectangular prism X and right rectangular prism Y?

Student-produced response: _____

2.32 adc9a053-ace5-4475-8499-b52461f5f6e1 · Hard

A right circular cone has a height of 22 centimeters cm and a base with a diameter of 18 cm. The volume of this cone is $n\pi$ cm^3. What is the value of n?

Student-produced response: _____

2.33 025459-DC · Hard

A cube has a surface area of 54 square meters. What is the volume, in cubic meters, of the cube?

A. 18

B. 27

C. 36

D. 81

2.34 9657587b-66c2-4743-91d3-22e3bf8ab94e · Hard

Triangles ABC and DEF are similar. Each side length of triangle ABC is 4 times the corresponding side length of triangle DEF. The area of triangle ABC is 270 square inches. What is the area, in square inches, of triangle DEF?

Student-produced response: _____

2.35 a5e0e74e-362f-4f21-aade-6a5d1b2518e0 · Hard

Rectangle $ABCD$ is similar to rectangle $EFGH$. The area of rectangle $ABCD$ is 648 square inches, and the area of rectangle $EFGH$ is 72 square inches. The length of the longest side of rectangle $ABCD$ is 36 inches. What is the length, in inches, of the longest side of rectangle $EFGH$?

A. 4

B. 9

C. 12

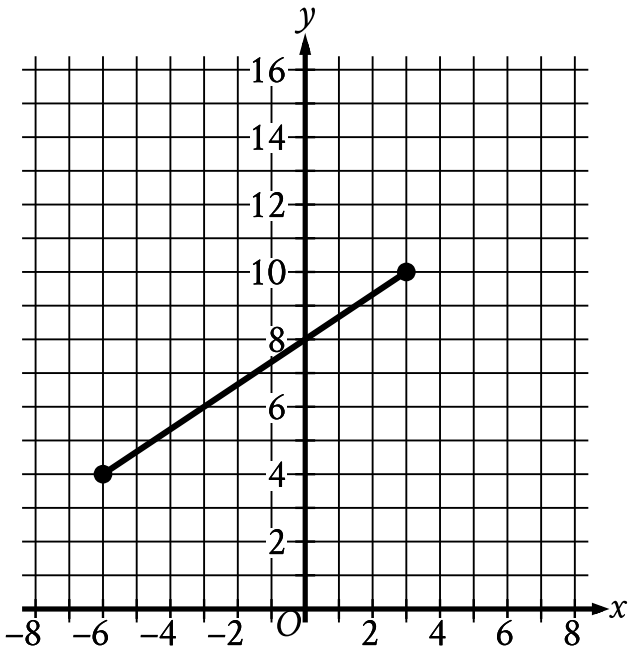
D. 36

2.36 243579cd-f8bb-4573-bce8-ea57314f3f36 · Hard

A right circular cone has a height of 22 centimeters cm and a base with a diameter of 6 cm. The volume of this cone is $n\pi$ cm^3. What is the value of n?

Student-produced response: _____

2.37 c682f49c-ee6e-439a-a3c9-64f6aab47176 · Hard



- The line segment slants gradually up from left to right.
- The line segment begins at the point (negative 6 comma 4).
- The line segment ends at the point (3 comma 10).

The line segment shown in the *xy*-plane represents one of the legs of a right triangle. The area of this triangle is $36\sqrt{13}$ square units. What is the length, in units, of the other leg of this triangle?

A. 12

B. 24

C. $3\sqrt{13}$

D. $18\sqrt{13}$

2.38 95dd10f0-9b55-4202-a756-91f1ede906ac · Hard

A right triangle has sides of length $2\sqrt{2}$, $6\sqrt{2}$, and $\sqrt{80}$ units. What is the area of the triangle, in square units?

A. $8\sqrt{2} + \sqrt{80}$

B. 12

C. $24\sqrt{80}$

D. 24

2.39 bca754b0-4853-461c-a71b-c465e54cbb57 · Hard

A right square prism has a height of 14 units. The volume of the prism is $2{,}016$ cubic units. What is the length, in units, of an edge of the base?

Student-produced response: _____

2.40 3b48c674-430c-4197-b876-ffb5199bc927 · Hard

Trapezoid A and trapezoid B are similar. The length of each side of trapezoid A is 41 times the length of the corresponding side of trapezoid B. The area of trapezoid A is how many times as large as the area of trapezoid B?

Student-produced response: _____

2.41 518073a0-4d7c-48ab-af96-3d1d19b33661 · Hard

Parallelogram *ABCD* is similar to parallelogram *PQRS*. The length of each side of parallelogram *PQRS* is 2 times the length of its corresponding side of parallelogram *ABCD*. The area of parallelogram *ABCD* is 5 square centimeters. What is the area, in square centimeters, of parallelogram *PQRS*?

A. 7

B. 10

C. 20

D. 25

2.42 01daee35-6a84-4baf-8ed5-3ef4ff55b251 · Hard

A cube has an edge length of 68 inches. A solid sphere with a radius of 34 inches is inside the cube, such that the sphere touches the center of each face of the cube. To the nearest cubic inch, what is the volume of the space in the cube not taken up by the sphere?

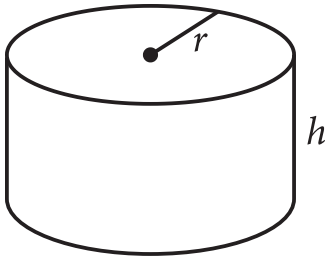
A. $149{,}796$

B. $164{,}500$

C. $190{,}955$

D. $310{,}800$

2.43 c1c7e4d7-d6dc-4e54-8b49-cbd88eab7153 · Hard



The figure shown is a right circular cylinder with a radius of r and height of h. A second right circular cylinder (not shown) has a volume that is 392 times as large as the volume of the cylinder shown. Which of the following could represent the radius R, in terms of r, and the height H, in terms of h, of the second cylinder?

A. $R = 8r$ and $H = 7h$

B. $R = 8r$ and $H = 49h$

C. $R = 7r$ and $H = 8h$

D. $R = 49r$ and $H = 8h$

2.44 06496-DC · Hard

A manufacturer determined that right cylindrical containers with a height that is 4 inches longer than the radius offer the optimal number of containers to be displayed on a shelf. Which of the following expresses the volume, V, in cubic inches, of such containers, where r is the radius, in inches?

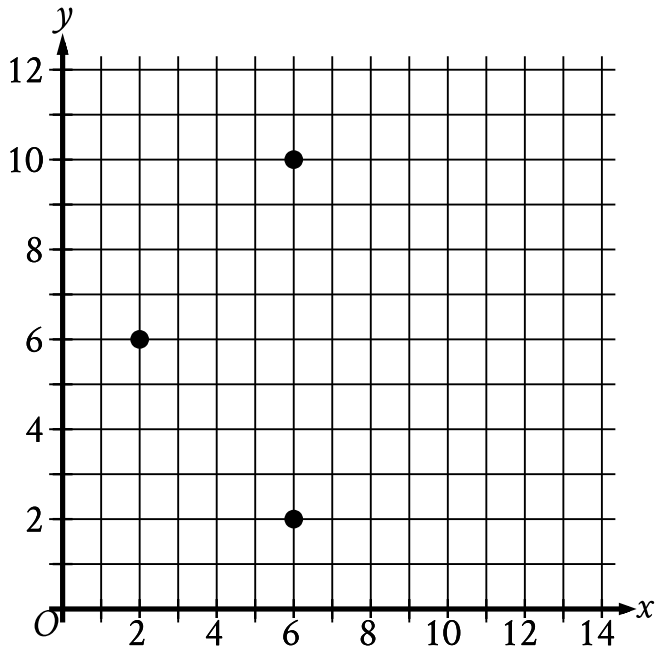
A. $V = 4\pi r^3$

B. $V = \pi(2r)^3$

C. $V = \pi r^2 + 4\pi r$

D. $V = \pi r^3 + 4\pi r^2$

2.45 e017d068-1e14-45a7-8981-963bc0f7fc9a · Hard



- From left to right, the 3 points have the following coordinates:
  - (2 comma 6)
  - (6 comma 10)
  - (6 comma 2)

The three points shown define a circle. The circumference of this circle is $k\pi$, where k is a constant. What is the value of k?

Student-produced response: _____

2.46 31c28a2a-1001-4c2c-b27f-aa2b2380c77d · Hard

A cube has a volume of 474,552 cubic units. What is the surface area, in square units, of the cube?

Student-produced response: _____

2.47 032655-DC · Hard

A right circular cone has a volume of $\frac{1}{3}\pi$ cubic feet and a height of 9 feet. What is the radius, in feet, of the base of the cone?

A. $\frac{1}{3}$

B. $\frac{1}{\sqrt{3}}$

C. $\sqrt{3}$

D. 3

2.48 1f5958dc-ffc1-426c-8358-d8fbb4354de4 · Hard

The circumference of the base of a right circular cylinder is 20π meters, and the height of the cylinder is 6 meters. What is the volume, in cubic meters, of the cylinder?

A. 60π

B. 120π

C. 600π

D. $2{,}400\pi$

2.49 555b2745-a82f-4231-a6df-c31085a80638 · Hard

Square A has side lengths that are 166 times the side lengths of square B. The area of square A is *k* times the area of square B. What is the value of *k*?

Student-produced response: _____

2.50 bb20b967-73fe-4c5e-97f8-e5887afb9aab · Hard

The edge length, in inches, of cube Y is $\frac{3}{88}$ the edge length, in inches, of cube X. The surface area, in square inches, of cube Y is *n* times the surface area, in square inches, of cube X. What is the value of *n*?

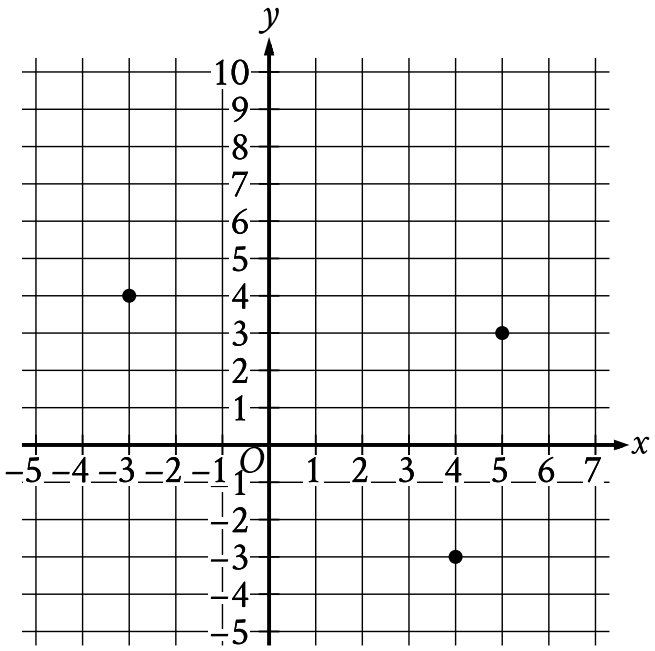
A. $\frac{9}{7,744}$

B. $\frac{27}{3,872}$

C. $\frac{3}{88}$

D. $\frac{9}{44}$

2.51 8f07857d-c0af-4793-9305-0e414671d257 · Hard



- Moving from left to right, the points have the following coordinates:
  - (negative 3 comma 4)
  - (4 comma negative 3)
  - (5 comma 3)

What is the area, in square units, of the triangle formed by connecting the three points shown?

Student-produced response: _____

2.52 b44ba6ac-80e2-4dec-ab99-a08180e2db81 · Hard

A right prism has a square base with side lengths of 10 inches. The height of the prism is 58 inches. What is the volume, in cubic inches, of the prism?

Student-produced response: _____

2.53 038033-DC · Hard

The dimensions of a right rectangular prism are 4 inches by 5 inches by 6 inches. What is the surface area, in square inches, of the prism?

A. 30

B. 74

C. 120

D. 148

2.54 5c9efc7d-a1db-4027-b9b5-a10ca2abedcd · Hard

Two identical rectangular prisms each have a height of 90 centimeters cm. The base of each prism is a square, and the surface area of each prism is K cm^2. If the prisms are glued together along a square base, the resulting prism has a surface area of $\frac{92}{47}K$ cm^2. What is the side length, in cm, of each square base?

A. 4

B. 8

C. 9

D. 16

2.55 64a7c0bf-065d-47a5-a938-4a1fabad6933 · Hard

Circle A has a radius of $3n$ and circle B has a radius of $129n$, where n is a positive constant. The area of circle B is how many times the area of circle A?

A. 43

B. 86

C. 129

D. $1{,}849$
